

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0146

Lake Charles Plant 100 kV
March 05, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0146. The Interconnection Customer proposes a 60.0 MW wind facility interconnecting at the Lake Charles Plant 100 kV substation in the Illinois area. The request was received on July 25, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 104 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,946,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Fort Hubug / Orchard Rd 500 KV, the Fort Humbug 500/345 kV Autotransformer loads to 121.1% of its 628 MVA rating. Project DFAX: 5.2%.

Piney Creek Project 500/345kV TX: 116.2% loading (776 MVA) following the outage of the Hilltop / Piney Creek Project Substation 500 KV; DFAX = 9.9%.

For the loss of Hilltop / Talenenergy Martins Creek Llc Williamsport 500 KV, the Hilltop 500-345 kV TX loads to 115.6% of its 806 MVA rating. Project DFAX: 8.1%.

Fort Humbug / W Campus 345 KV: 113.9% loading (1,371 MVA) following the outage of the Ft Humbug / Orchard Road Substation 500 KV; DFAX = 9.1%.

For the loss of Boca Raton Substation / Pueblo Xtra 500 KV, the Pueblo Xtra 500-230 kV Autotransformer loads to 113.8% of its 766 MVA rating. Project DFAX: 46.2%.

Talenenergy Martins Crk Llc Williamsport 500 kV/345 kV Transformer Bank: 112.7% loading (821 MVA) following the outage of the Hilltop 500 kV/345 kV Transformer; DFAX = 7.0%.

The Little Chute / Talenenergy Martins Creek Llc Harrisburg Substation 345 KV Line is overloaded to 112.5% of its 1,012 MVA emergency rating for the loss of the Coronet / Ft Humug 500 KV. The project contributes 7.6% DFAX (4.5 MW) to this overload.

For the loss of Beaver / Trego 500 KV, the Adams / Onancock Diesel Generation Facility Substation 230 KV loads to 112.3% of its 670 MVA rating. Project DFAX: 17.3%.

The Norwich / Larson Street 230 KV line is overloaded to 103.9% of its 559 MVA emergency rating for the loss of the Barrett / Norwich 500 KV. The project contributes 9.7% DFAX (5.8 MW) to this overload.

The American / Urban 345 KV is overloaded to 103.5% of its 1,270 MVA emergency rating for the loss of the West Waynesboro / Chambersburg 345 KV Line. The project contributes 5.2% DFAX (3.1 MW) to this overload.

Carlyss / Boca Raton 230 KV line: 102.8% loading (560 MVA) following the outage of the Carlyss Substation / Top Crop 500 KV; DFAX = 5.7%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fort Humbug 500/345 kV Autotransformer	Replace transformer with higher-rated unit	\$31,298,000	43% (\$13,413,000)
Piney Creek Project 500/345kV TX	Replace transformer with higher-rated unit	\$22,447,000	1% (\$256,000)
Hilltop 500/345 kV Auto	Replace transformer with higher-rated unit	\$30,084,000	1% (\$346,000)
Ft Hmbug / West Campus 345 KV	Rebuild line with higher-capacity conductor	\$42,833,000	1% (\$495,000)
Pueblo Xtra 500-230 kV Autotransformer	Replace transformer with higher-rated unit	\$23,526,000	2% (\$370,000)
Talenenergy Martins Crk Llc Williamsport 500 kV/345 kV XFMR	Replace transformer with higher-rated unit	\$20,412,000	1% (\$236,000)
Little Chute / Talenenergy Martins Creek Llc Harrisburg Substation 345 KV Line	Rebuild line with higher-capacity conductor	\$16,219,000	1% (\$187,000)
Adams / Onancock Diesel Generation Facility 230 KV	Rebuild line with higher-capacity conductor	\$34,748,000	1% (\$396,000)
Norwich / Larson Street 230 KV line	Reconductor line	\$6,479,000	1% (\$69,000)
American / Urban 345 KV	Reconductor line	\$22,607,000	43% (\$9,689,000)
Carlyss / Boca Raton 230 KV line	Reconductor line	\$12,677,000	12% (\$1,489,000)
	Total allocated to project		\$26,946,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.